

Answer: A

17 rate of increase in internal energy = rate of heating the filament + rate of doing work on filament by the gas

0 (since temperature is constant) = -20 (rate of heating since the temperature is not rising, it is *losing* heat) + rate of doing work on filament by the gas

Rate of doing work on filament by the gas = +20 W

Answer: D

18
$$\frac{(1.0 \times 10^5) \text{ V}}{(25 - 270) \text{ V}} = \frac{P_{\text{new}} 2\text{V}}{(25 - 270) \text{ V}}$$